

BOCES Service Ticket Requests: Broome-Tioga & Maine-Endwell

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Executive Summary

Over the course of the year, schools can encounter a range of challenges that hinder learning, from inadequate program funding to limited resources for students. In New York State, the Boards of Cooperative Educational Services (BOCES) plays a key role in addressing these challenges by providing shared educational and technological services to school districts. Through Regional Information Centers (RICs), BOCES supports districts by managing technology infrastructure, instructional programs, and operational systems. One such support system is the ServiceNow portal, which allows teachers, students, or even parents to submit service tickets for assistance.

By identifying obstructions within the ticket resolution process for the Broome-Tioga and Maine-Endwell school districts, issues submitted can be resolved more efficiently and in a timely manner in the future. Thus, a stronger learning environment is created.

From performing data analysis on the service ticket data for the Broome-Tioga and Maine-Endwell school districts, we identified significant bottlenecks in the form of excessive ticket transfers between assignment groups, as well as seasonal spikes in ticket volume that contributed to resolution delays. However, our findings also revealed several operational strengths, including fast initial response times, high resolution rates within teams, and increasingly clear assignment group roles. By analyzing patterns and comparing them with the previous groups' analyses, we proposed targeted recommendations, such as implementing summer or winter operations plans, creating a redirection cheat sheet, and more.

Our work with the BOCES helps provide them with actionable insights to reduce inefficiencies, maintain momentum on recent improvements, and ensure timely technical support across the academic year.

Introduction

The Boards of Cooperative Educational Services, otherwise known as BOCES, were established by the New York State Legislature in 1948 as a means to facilitate collaboration among local school districts, especially in rural areas, by providing shared educational services while reducing costs for families (BOCES: A History, n.d.). Over the past 75 years, BOCES has adapted to the ever-changing needs of students and schools across the state. Today, New York has 37 BOCES, each of which functions as a dynamic center for educational innovation, continually expanding and adapting to enhance learning opportunities for all students (About BOCES, n.d.). Among them is Broome-Tioga BOCES, which aims to serve the Southern Tier Region of New York State by helping public schools control costs and providing quality

programs by sharing services. To provide these services to the Southern Tier Region, Broome-Tioga BOCES works closely with the SRIC (South Central Regional Information Center), a part of BOCES that provides technological assistance to ensure that school districts can gain access to the services the program offers. Our paper and analysis will focus on the Broome and Maine-Endwell school districts.

There is a growing need to assess how service operations have changed over time and where improvements can be made. With increased reliance on technology in educational settings, ensuring timely and effective IT support is more important than ever. Our analysis aims to evaluate how efficiently service tickets are resolved across districts and assignment groups, identify operational bottlenecks, and uncover patterns in ticket reassignment and resolution time. By comparing our findings with those of the previous team, who worked on a similar dataset in the past, we also aim to evaluate the progress and provide updated insights that can inform future decision-making. Improving ticketing efficiency directly impacts the quality and timeliness of support that students, teachers, and staff receive, ultimately enhancing productivity and reducing disruptions within BOCES districts. Understanding these operational patterns allows administrators to strategically allocate resources, streamline processes, and foster a more responsive, effective support environment.

To do this, we performed data analysis on the service ticketing data through Power BI for this report. Because our project is a continuation of a previous group's work from last year, we recreated visualizations from their report to enable direct comparisons. By comparing graphs from two datasets covering different time periods, we can see if any progress or improvements have been made since the last report. We also looked for other bottlenecks that have not been identified and looked for ways to improve the ticketing resolution process.

Methods

Dataset

Our dataset contains service tickets collected from two BOCES regions: Broome-Tioga BOCES and Maine-Endwell BOCES. The Broome-Tioga dataset contains tickets resolved between May 2022 and February 2024, while the Maine-Endwell dataset covers tickets between August 2023 and December 2024. To support our analysis, it is important to outline how tickets are created, processed, and resolved.

Tickets are categorized as either *incidents* (INC) that indicate a malfunction or technical issue or *requests* (TASK) that involve a service or resource need. Service tickets are created through two primary channels. District employees can submit tickets directly through the ServiceNow platform. On the other hand, students, parents, and teachers can also contact the Service Desk by phone, where an employee can create a ServiceNow ticket on their behalf.

Upon creation, each ticket is assigned to an assignment group responsible for addressing the issue. Assignment groups are specialized groups that can work on a specific set of tickets, such as *Field Services*, *Application Support & Training*, and *TAC Tier I*. For example, the TAC

Tier I group is the Service Desk at the district level and is typically the first point of contact for those who submit tickets.

If the Service Desk is able to resolve the issue at the point of contact, the ticket is closed immediately. Otherwise, it is forwarded to the appropriate group based on the nature of the issue. The initial group assignment is determined by the individual who created the ticket. Should a ticket require expertise outside the assignment groups' scope, or if the ticket was misassigned, it can be transferred to another group. In our dataset, we can see the number of ticket transfers when there is a repeated ticket number. Ultimately, the final group to work on the ticket is responsible for resolving and closing it.

An example of an artificial row is depicted below in Table 1. The dataset is structured such that each row represents work completed by a specific assignment group. Consequently, tickets that are transferred between multiple groups appear in multiple rows. For example, if INC12345 is seen four times in the dataset, it would indicate that the ticket was transferred three times. The first time would not count as a transfer because it would be the initial submission of the ticket.

	Artificial Example	Description
Number	INC12345, TASK23456	Ticket number assigned. “INC” indicates an Incident, while “TASK” indicates a Request
Opened	10/23/2024 07:59:49 AM	Date and time the ticket was first opened
Closed	11/07/2024 03:35:03 PM	Date and time the ticket was first closed
Short Description	Reset Password - John Doe...	Purpose of the ticket
Service Offering	End User Device Service	What type of service was done
District	Maine-Endwell CSD	What district was it submitted for
Building	Maine-Endwell High School	What building was it for
Assignment Group	TAC Tier II	Which groups worked on it
Start	12/17/2024 03:30:30 PM	When the group first got the ticket
End	12/16/2024 09:03:00 AM	When that group finished working on it
Technical Notes	02/15/2024 09:26:25 AM - John Doe (Technical Notes) reset password...	Communication between technical support staff
Comments To/From Customer	02/14/2024 07:40:59 AM - John Doe (Comments To/From Customer) Good morning...	Communication between staff and customers
Priority	4 - Low	Level of priority
Close Code	Configuration Change	Action the assignment group took to solve the issue

Table 1. A table with artificial examples of rows in the dataset.

This structure is critical to understanding the cycle and complexity of service ticket resolutions. It enables detailed analysis of reassignment patterns, workload distribution across groups, and the efficiency of the ticket resolution process within and across districts.

Cleaning the Dataset

Before conducting any analysis on our dataset, we went through several phases of data preparation and cleaning to ensure the dataset was ready for meaningful interpretation. While we did not modify the original dataset itself, we created several calculated columns in Power BI to enhance the quality and depth of the analysis.

These new columns allowed us to measure performance, identify patterns, and filter results in more intuitive ways. The following fields were created as part of this preparation process:

- *Duration Open*: Calculated the number of days a ticket remained open, based on its open and close dates
- *Month Year Sorted*: A custom field used to sort visuals chronologically for clearer trend analysis
- *Redirection Category*: Classified tickets based on the number of times they were transferred between assignment groups
- *Redirection Count*: Captured how many times a ticket was redirected (number of entries per ticket minus one)
- *Average Open Time*: Used to calculate the average duration tickets remained open across assignment groups or time periods
- *Ticket Type*: Consolidated categories to differentiate between incidents and requests
- *Number of Transfers*: Similar to redirection count, but formatted for easier display in visualizations
- *Ticket Count*: Enabled aggregation of ticket volume by group, data, or category

These derived columns helped us avoid altering the raw data while still giving us the necessary tools to explore relationships, identify inefficiencies, and track performance over time.

Creating Visualizations

A key objective of this analysis was to compare current service ticket data from Broome-Tioga and Maine-Endwell BOCES with prior analysis conducted by a previous group that worked on past data of the same nature. To achieve this, we recreated several of their visualizations using the updated dataset, allowing for a direct comparison of trends, performance, and structural changes over time. This approach allowed us to assess how operation patterns, assignment group activity, and ticket handling efficiency have evolved over time.

In addition to replicating key visualizations from the previous analysis, we also developed new visualizations to address areas that were not covered in the previous work. These additions allowed us to explore new questions and insights, offering a more comprehensive understanding of current service operations.

We developed a series of interactive dashboards in Power BI that allowed us to filter the data by assignment group, ticket type, district, and time. This interactive structure made it easier to isolate specific patterns, such as how different teams handled incoming tickets and how these patterns evolved over time.

Several visualization methods were employed to support this analysis. Bar and line charts were used to track the number of tickets over time, making it possible to track shifts in service demand. They were also used to map the flow of tickets between assignment groups, revealing how often tickets were transferred. Pie charts were used to show the distribution of ticket types and to visualize the proportion of tickets that underwent one or more transfers, which helped us assess the overall efficiency of the process.

Results

Ticket Transfer Analysis

Analysis of ticket transfer activity identified clear patterns regarding which assignment groups handled the greatest number of ticket transfers within each of the BOCES districts, which can be seen in Figure 1 and Figure 2.

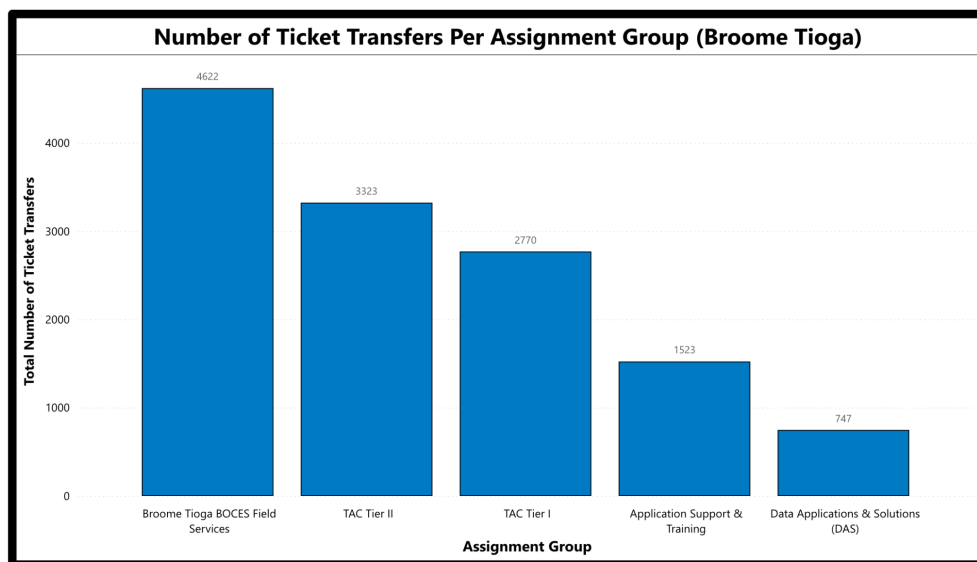


Figure 1. This figure shows the five assignment groups with the highest total number of ticket transfers in the Broome-Tioga District.

As shown in Figure 1 above, the assignment groups with the highest total number of ticket transfers were the Field Services team, with 4,622 transfers. The Broome-Tioga BOCES Field Services notably experienced significantly more transfers compared to other groups, suggesting that this team frequently a key factor for ticket redistribution and escalation.

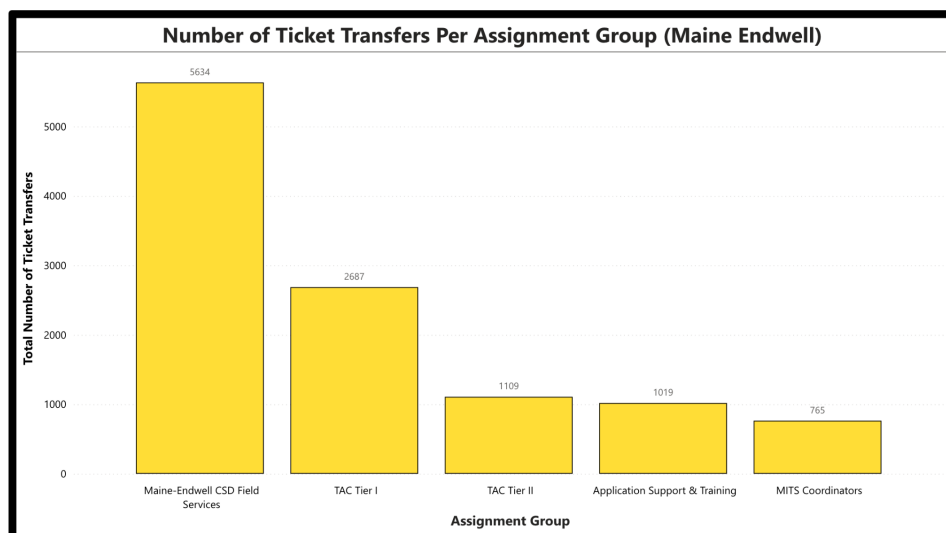


Figure 2. This figure shows the five assignment groups with the highest total number of ticket transfers in the Maine-Endwell District.

Similar to the Broome-Tioga District, Figure 2 above shows that their Field Service team is also the leading group for ticket transfers, with 5,634 transfers. This indicates that the Field Services team also serves as a central hub in the ticketing workflow.

A likely explanation for this could be that because the Field Services team covers a wide scope, they may receive issues that should have gone directly to more specific groups. Another possibility is that they may have more transfers because they simply receive tickets at a higher volume than any other assignment group. They may also receive a ticket that could require collaboration with other groups, leading to more transfers per ticket.

Ticket Resolution Analysis

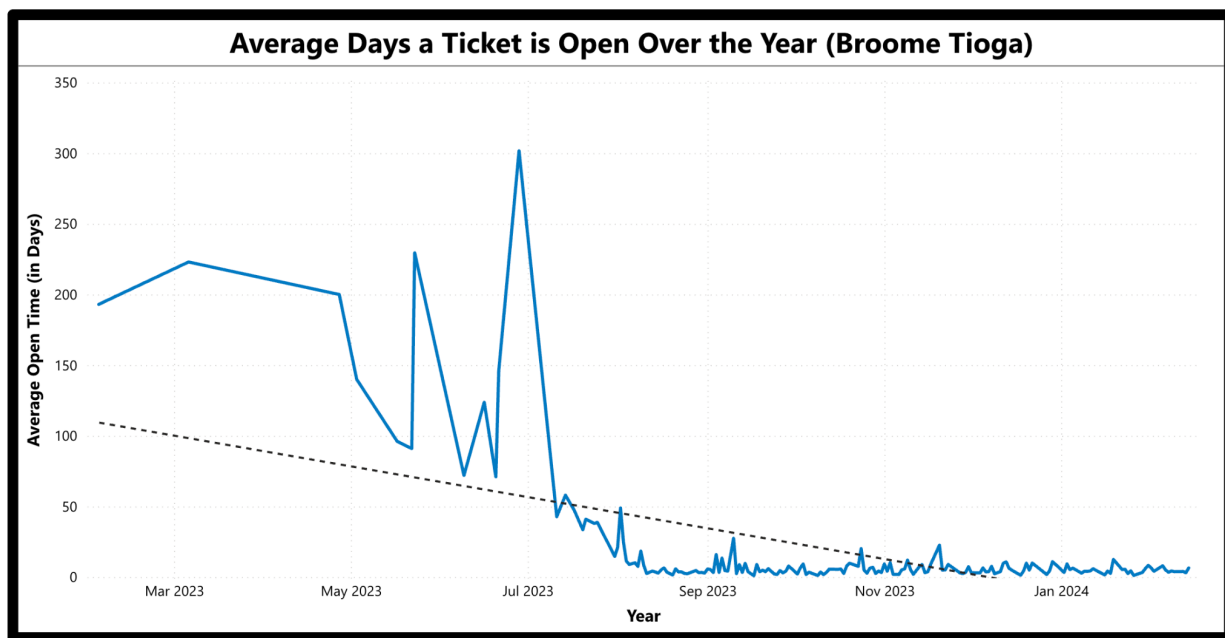


Figure 3. Line graph that shows the average number of days a ticket remained open in Broome Tioga over the year, highlighting a sharp decline in open times after July 2023 and an overall downward trend.

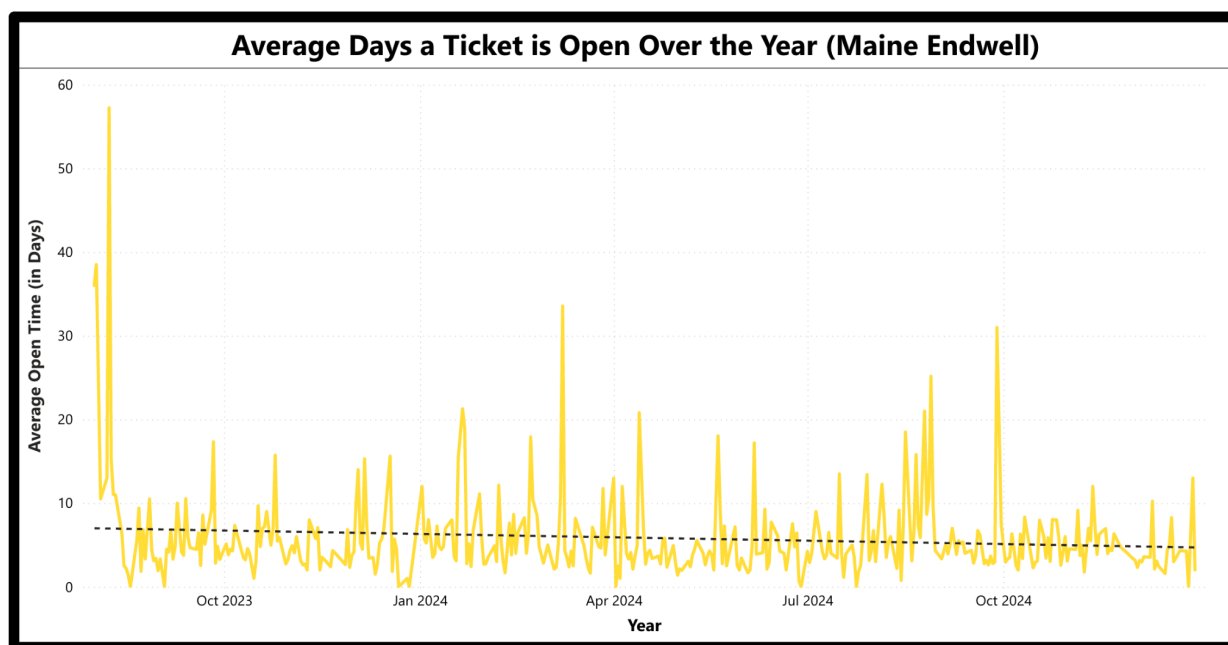


Figure 4. Line graph that shows the average number of days a ticket remained open over the year in Maine-Endwell, with fluctuating spikes but a generally decreasing trend in open time, indicated by the downward-sloping trendline.

Figures 3 and 4 above show the average open time of tickets by month for two different regions, Broome-Tioga (Figure 3) and Maine-Endwell (Figure 4). In Broome-Tioga, the average open time peaked in July 2023, reaching over 300 minutes, indicating significant delays in ticket resolution during that period. This spike may be due to a lack of staff availability during the summer months, as school is not in session and many employees might be on leave or working reduced hours. After July, there is a sharp and sustained decrease, with resolution times stabilizing to under 50 minutes by early 2024, suggesting operational improvements or increased staffing.

On the other hand, Maine-Endwell shows a more stable pattern, with average open times consistently fluctuating between 5 and 20 minutes. While occasional spikes occur, the fluctuations are less pronounced than those seen in the Broome-Tioga community. This stability may stem from Maine-Endwell's smaller size, as it is a subset of the broader Broome-Tioga region. These trends help answer the question about how long tickets remained open and what may have caused delays. While Maine-Endwell maintained consistent resolution times, possibly due to steady staffing or fewer tickets, Broome-Tioga may have experienced seasonal staffing issues that impacted resolution efficiency, especially in the summer months.

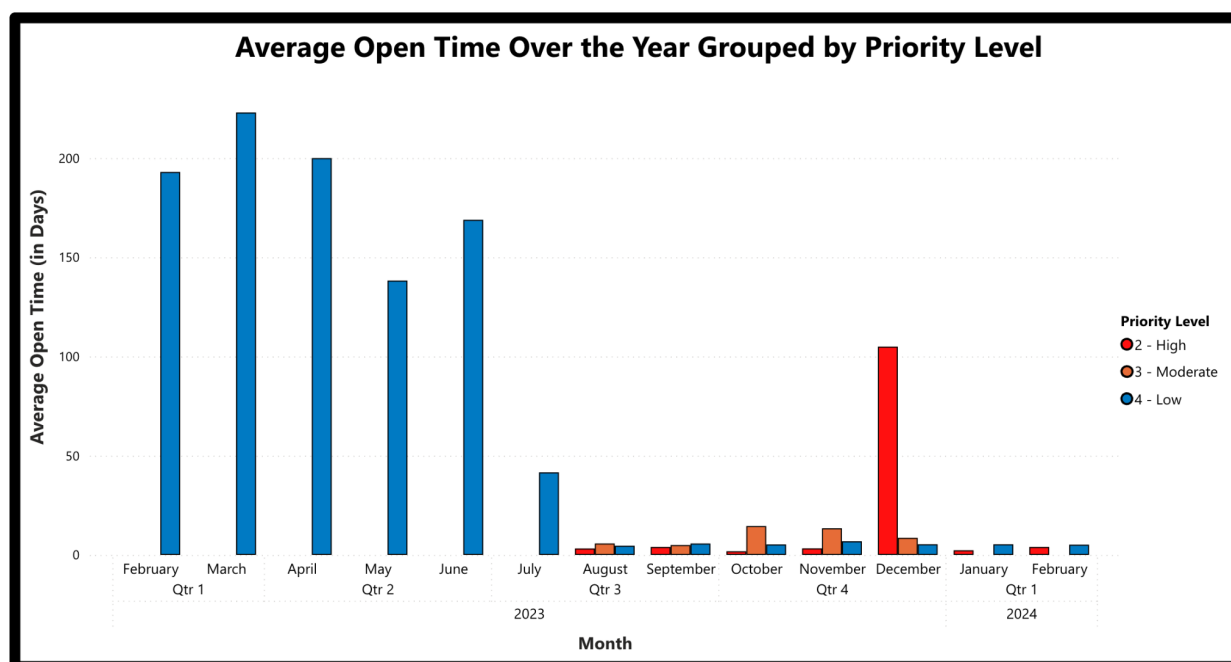


Figure 5. Bar graph that shows the average open time (in days) for tickets over the years, grouped by priority level (High, Moderate, Low), with a noticeable decline in low-priority open times after July 2023 and a spike in high-priority issues in December.

Figure 5 above shows the average open time (in minutes) for tickets in the Broome-Tioga region across various months, categorized by priority levels. A clear trend emerges where low-priority tickets (Priority Level 4, shown in blue) consistently experience significantly longer resolution times, especially during the first two quarters of 2023. For instance, from February to June, low-priority tickets had average open times ranging from 138 to 223 minutes. In contrast, moderate (Level 3) and high-priority (Level 2) tickets show considerably shorter open times throughout the rest of the year. In addition, Quarters 1 and 2 are almost entirely made up of low-priority tickets, while high-priority tickets only begin to appear in Quarter 4, with a spike in December, where a high-priority ticket had an average open time of 105 minutes. This suggests that the delay in ticket resolution may be partly attributed to the overwhelming number of low-priority tickets early in the year, which likely received less urgent attention. As higher-priority tickets started to emerge later in the year, their faster resolution times reflect the natural prioritization of more urgent issues. Overall, the distribution of ticket priority over time appears to be a key factor influencing average open times.

Comparisons with Previous Work

INC (“Incident”) Tickets

To evaluate improvements in ticket routing efficiency, we compared current INC ticket redirection data with findings from a previous analysis conducted by the previous group

members (as seen in Figure 6 and Figure 7). The comparison focuses on how frequently tickets were redirected to another group before the ticket was resolved.

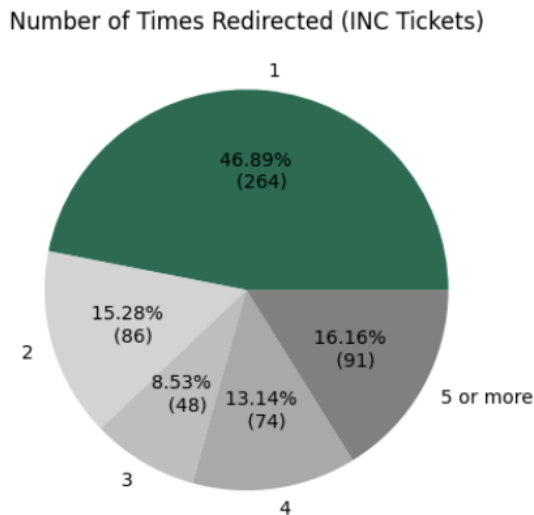


Figure 6. Pie chart displaying the frequency of ticket redirection between INC tickets. As you can see, most tickets are redirected to another group. This pie chart was taken from the report of the previous group (Michals & Stewart).

In the previous dataset, which included INC ticket data from three different BOCES districts, only 46.89% of tickets were redirected once, while a notable 38% of tickets were redirected three or more times (Figure 6). More notably, 16.16% of tickets were redirected five or more times, which highlights the inefficiencies in the initial ticket assignment process.

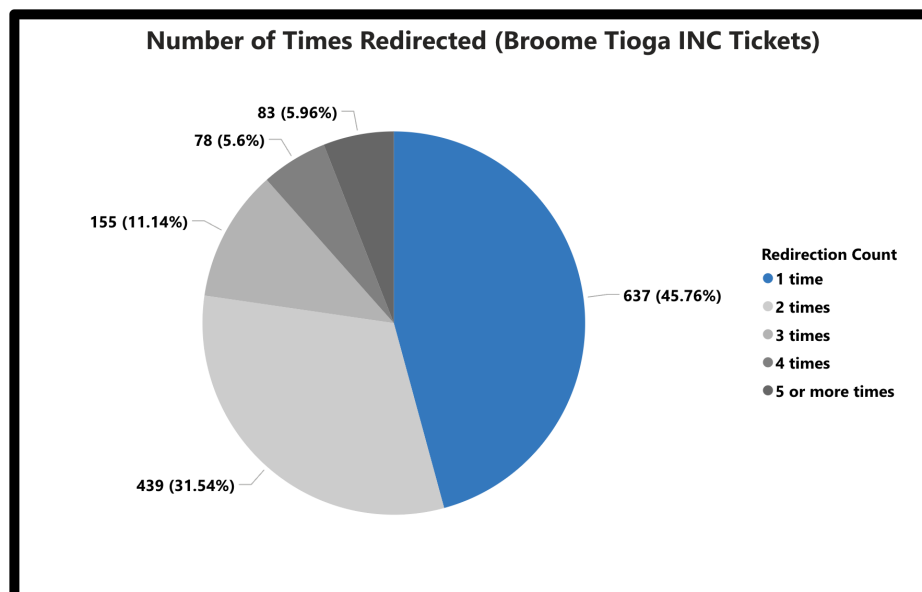


Figure 7. Pie chart adapted from that of the previous group with the current data for the INC tickets. As you can see, most tickets are redirected once.

In contrast, the current dataset, focused exclusively on the Broome-Tioga district, reveals that 45.76% of tickets were redirected once, maintaining a similar baseline (Figure 7). However, the proportion of tickets redirected three or more times fell to approximately 23%, and those redirected five or more times dropped substantially to just 5.96%, which represents a meaningful reduction of over 10% in excessive redirections.

Despite the broader scope of the current data, encompassing all of Broome-Tioga BOCES rather than select districts within it, the sharp decline in excessive redirections strongly indicates a meaningful improvement in ticket routing within Broome-Tioga as a whole. This improvement may reflect enhanced categorization at intake, more accurate routing protocols, or clearer assignment group definitions.

TASK (“Request”) Tickets

Number of Times Redirected (TASK Tickets)

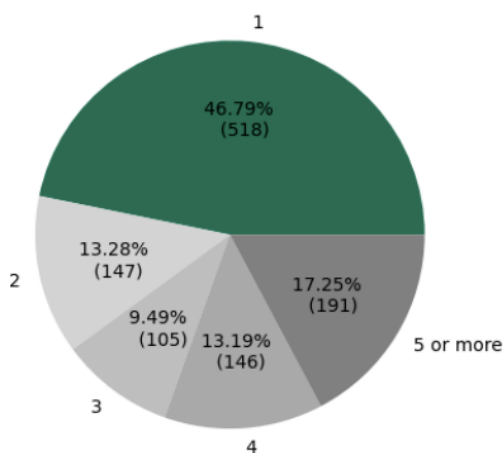


Figure 8. Pie chart displaying the frequency of ticket redirection between TASK tickets. Similar to the INC tickets, most tickets are redirected to another group. This pie chart was taken from the report of the previous group (Michals & Stewart).

A similar trend is evident in the analysis of the TASK tickets. In the previous data, only 46.79% of TASK tickets were redirected once, while the remaining majority experienced multiple handoffs (Figure 8 above). Specifically, 17.25% of tickets were redirected five or more times, and over half of the tickets required two or more redirections. This pattern indicates considerable inefficiency and likely confusion in initial ticket routing.

In the current Broome-Tioga TASK data (Figure 9 below), however, a significant shift is observed:

- (1) 89.6% of tickets were resolved after just one redirection;
- (2) 5.56% of tickets required two redirections; and
- (3) Only 2.87% reached three or more redirections.

This dramatic reduction in excessive ticket transfers for TASK tickets points to a clear improvement in internal workflow, better categorization at intake, and more consistent assignment practices¹.

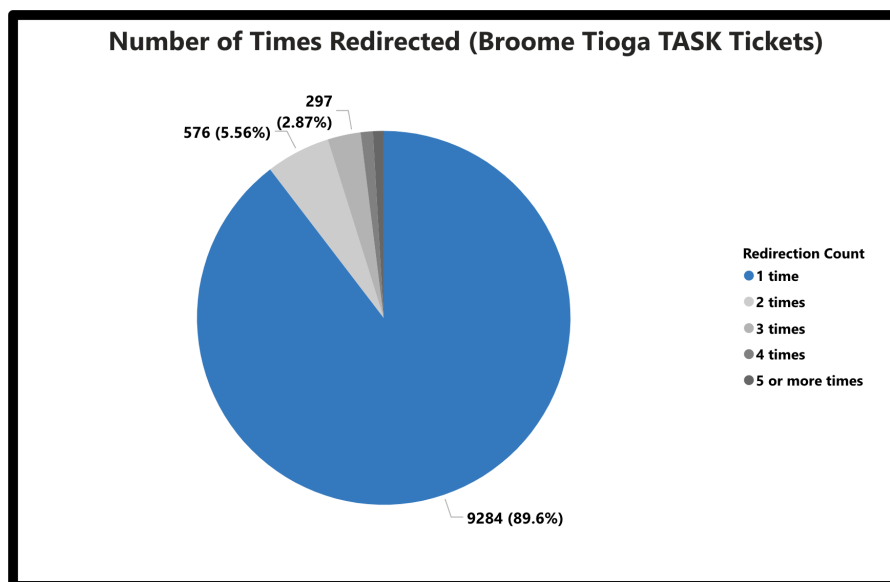
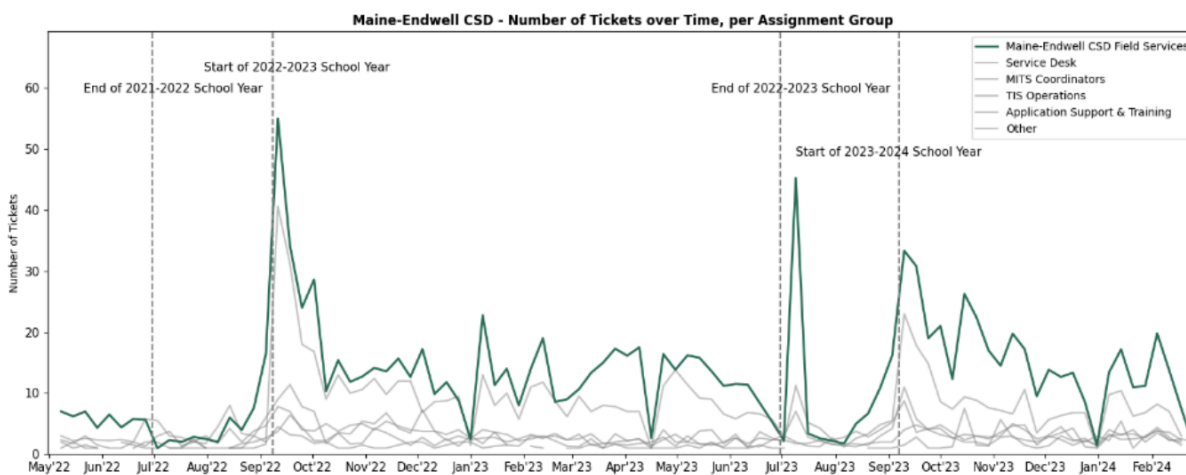


Figure 9. Pie chart adapted from that of the previous group with the current data for the TASK tickets. As you can see, most tickets are redirected once.

Ticket Counts: Maine-Endwell

To evaluate changes in service demand and assignment group workload over time, we also analyzed the number of tickets submitted in Maine-Endwell across two time periods using longitudinal graphs (Figures 10 and 11). Figure 10, created by the previous group, covers ticket volume from May 2022 to February 2024, while Figure 11, created as part of our current analysis, displays data from August 2023 to December 2024.



¹ There is additional analysis of the INC and TASK tickets for the Maine-Endwell district in Appendix A.

Figure 10. Line graph showing the number of tickets over the school year at the Maine-Endwell school district. Each line is representative of the assignment group. It can be seen that the peak ticket counts are at the beginning and end of the school year. This line graph was taken from the report of the previous group (Michals & Stewart).

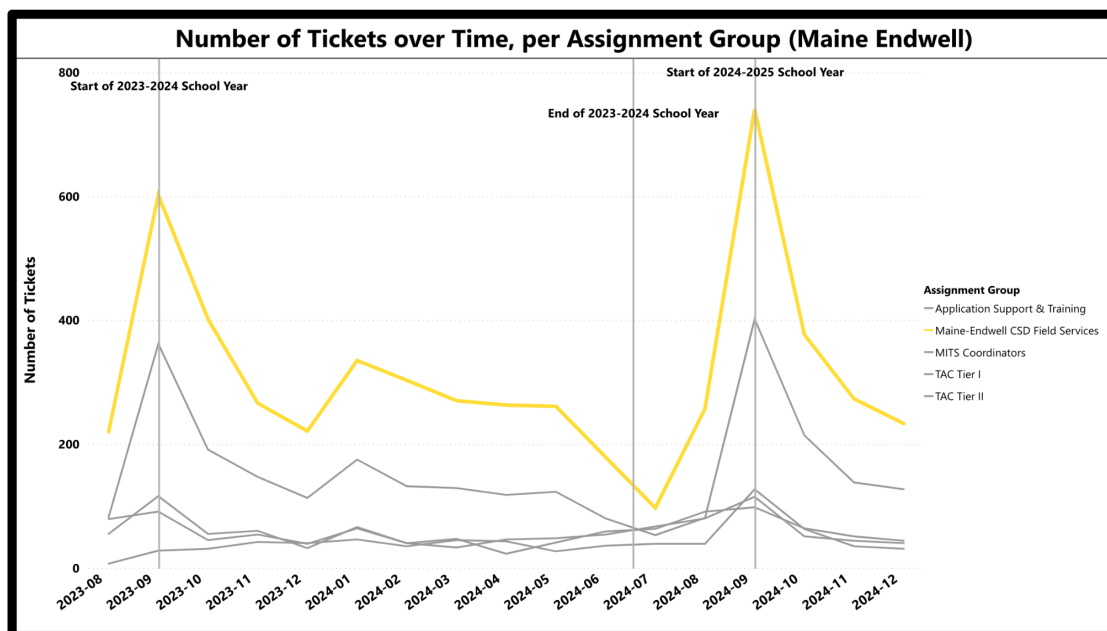


Figure 11. Line graph showing the number of tickets over the most recent school year at the Maine-Endwell school district. Similar to the previous school year, there are clear peaks in ticket counts at the beginning and end of the school year.

Figure 10, shown above, displays trends across the end of the 2021-2022 school year through the beginning of the 2023-2024 school year. In this period, clear spikes in ticket volume are observed at the start of each academic year, especially in September 2022. These surges are primary tickets from the Maine-Endwell Field Services group, which consistently handles the highest number of tickets.

Figure 11, shown above, covers the start of the 2023-2024 school year and that of the 2024-2025 school year. While the seasonal spike pattern continues, the magnitude of the ticket volume has increased significantly. In September 2024, the Maine-Endwell Field Services group handled over 700 tickets, compared to monthly peaks of around 50-60 in the early data. Additionally, TAC Tier I and Tier II appear more prominently, which suggests that there could have been an increase in workload transparency or clearer role delineation across the different groups.

These recurring surges at the start and end of each school year are consistent with expected academic demands. It is understandable that at the beginning of the school year, demands rise because all the students, teachers, and staff are returning to school for the first time since the previous school year. After a long summer break, the beginning of the year likely calls

for any leftover issues to be resolved or issues to arise. Examples of this could be due to device distribution, account setup, onboarding, or even technology setups in classrooms. These explanations also align similarly with the end of the school year because everything is winding down, and there are likely more tickets for issues that were going on all school year that students, teachers, and staff held off on resolving. With the long summer break approaching, these spikes could be due to the latter aspect, such as device returns, staff offboarding, final grade submissions, and infrastructure reviews.

The overall increase in ticket volume is a very interesting result of this analysis. There could be several factors that contribute to the ticket increase. While this could be explained by a greater use of classroom technology, a larger user base, or a greater adherence to submitting tickets rather than relying on informal support systems, it is also possible that overlapping data from the previous academic year could have contributed to the elevated ticket counts in our graph. For instance, if the earlier months of our dataset captured some tickets from the tail end of the 2022-2023 school year, which is partially represented in the previous group's report, this could artificially inflate the ticket counts.

Despite the increase and this potential overlap, there are clear signs of workflow improvements. It is evident that the ticket load is more evenly distributed across teams compared to earlier data. We can also see that there is increased visibility of TAC Tier I and II, which implies that routing accuracy may have improved as well. By comparing the two figures, there is also evidence that shows that ticket surges stabilize more quickly now, which shows that there is more efficient handling capacity².

Conclusion

Our analysis of BOCES service ticket data across the Broome-Tioga and Maine-Endwell regions has revealed key patterns related to ticket transfers, resolution times, redirection trends, and workload distribution within BOCES. Field Services teams consistently had the highest number of ticket transfers in both districts, highlighting their role as central hubs in the support workflow at BOCES. While Broome-Tioga experienced a spike in ticket open times during July 2023, likely because of staffing shortages during the summer months, resolution times dropped sharply afterwards, which indicated improvements in the process. In contrast, Maine-Endwell maintained relatively consistent average open times throughout the year, with occasional but manageable spikes. This comparison highlights potential operational or structural differences between the two regions that influence service performance. Comparisons with previous group findings also show a notable decrease in excessive ticket redirections, which shows improvements in intake and routing practices. Despite a rise in overall ticket volume in Maine-Endwell, support teams appear to have adapted well, distributing the workload more evenly and responding to seasonal spikes more efficiently.

² Another comparison with the previous group's work can be seen in Appendix B. This graph did not show a significant difference in results, so it was not included in this portion of the paper.

These findings are important for our community partner as they provide clear evidence that resource allocation and staffing availability directly affect service quality. Understanding the seasonal patterns in Broome-Tioga can help administrators anticipate slowdowns and proactively implement solutions, such as temporary staffing or adjusted workflows during school breaks. For Maine-Endwell, the stable trend suggests that current systems may be effectively supporting their ticket resolution process, offering a potential model for Broome-Tioga to explore or replicate.

Limitations and Future Research

While this analysis shows important performance trends, there are still many limitations. Our evaluation focuses mainly on quantitative data and relies on monthly data aggregation, which may obscure short-term fluctuations or delays. It also does not account for ticket complexity, resolution quality, or user satisfaction, which are all important components in evaluating overall service effectiveness. Additionally, differences in how tickets were logged and categorized over time may have influenced the comparability of current and past data.

A significant limitation was the lack of access to the original code used by the previous group. This was simply due to data privacy issues, where the previous group's visualizations were created through Google Colab, which lacked the data security that the BOCES required. Therefore, we were only provided with their final report and poster, which had visualizations but lacked the precise data cleaning and visualization code. This required us to manually reconstruct their visualizations for comparison, a process that introduced uncertainty, especially given that accurate replication was a major priority for the BOCES team. While we were able to replicate the majority of the visualizations, access to their original code or processing steps could have ensured more consistent and precise comparative analysis.

Another limitation lies in the differences regarding the tools used for visualization: our visualizations were built with Power BI, whereas the previous group's visualizations were built with Google Colab, as previously discussed. These tools differ significantly in their flexibility, customization, and data processing capabilities, which may have led to discrepancies in how metrics were calculated and displayed.

Future research could involve qualitative insights from IT staff or users to better understand process delays in the system. Additionally, comparing these findings with similar organizations outside the BOCES community could help identify broader trends and best practices in educational IT support.

By addressing these limitations, future work can build on this foundation to improve service processes, enhance routing efficiency, and ensure timely support across all stages of the academic calendar.

Policy Recommendations

In terms of policy implications, we recommend a couple of different tactics to improve the workflow for the BOCES in terms of service request tickets:

- Implementing a summer or winter operations plan that includes flexible staffing models
- Creating a redirection cheat sheet that lists common issues and which teams they would go to for easier and more efficient ticket rerouting to reduce more than one redirection
- Using automation tools to reduce delays during predictable low-staff periods due to breaks
- Providing additional basic troubleshooting training for the Service Desk staff, so that they could resolve more tickets without escalating
- Having priority level escalate depending on *X* days open, so that the higher priority tickets can be reviewed more quickly
- Implementing a transfer ticket alert that will notify a supervisor if a ticket is transferred more than two times to prevent repeated misrouting

By utilizing these insights, BOCES can strengthen its support systems and ensure that students, teachers, and staff receive timely technical assistance throughout the entire school year.

Operational Highlights

In the midst of identifying areas of improvement, it is equally important to recognize the strengths already present within the BOCES. Our analysis revealed several operational highlights that demonstrate the team's commitment to effective service delivery:

- *Fast Initial Responses*: The visualizations suggest that tickets are consistently acknowledged shortly after submission, which indicates strong responsiveness and attention to incoming requests.
- *High Resolution Rates within Groups*: Many tickets are resolved without being transferred to other groups and have improved significantly since the previous analysis, which reflects confidence among support staff in addressing issues at the first point of contact.
- *Clear Assignment Group Roles*: The distribution of tickets across a range of assignment groups, particularly the growing presence of TAC Tier I and II, shows that responsibilities are becoming more defined. This clarity in the roles of each group helps to reduce confusion in routing and ensures issues are being directed to the appropriate groups.

Recognizing these strengths not only affirms the team's progress over the past year but also provides a practical framework for scaling successful strategies across the BOCES team.

Works Cited

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Appendix A

INC and TASK Tickets for Maine-Endwell

This appendix presents additional analysis on ticket redirection trends for the Maine-Endwell district, separated by ticket type, either INC (Incident) or TASK (Request). These pie charts provide further insight into how efficiently tickets are routed and resolved within the district's support system.

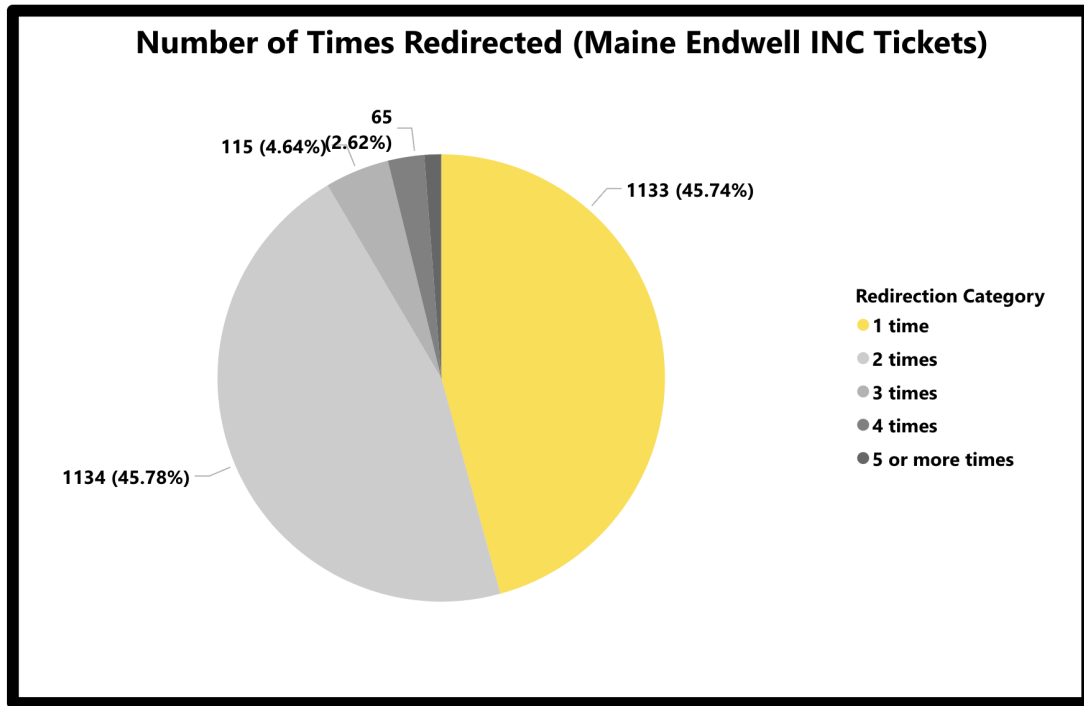


Figure A1. Pie chart adapted from that of the previous group with the current data for the INC tickets for the Maine-Endwell district. As you can see, most tickets are redirected twice by only 0.04%.

Figure A1 displays the redirection frequencies for INC tickets. The majority of these tickets were redirected twice, unlike all the other redirection analyses. However, the majority is only by a mere 0.04%, with the close second being one time. This near-even distribution suggests that while many tickets are resolved after the first reassignment, a substantial proportion still require a second redirection, which points to areas where initial routing accuracy could be improved. Only a small percentage required three or more redirections.

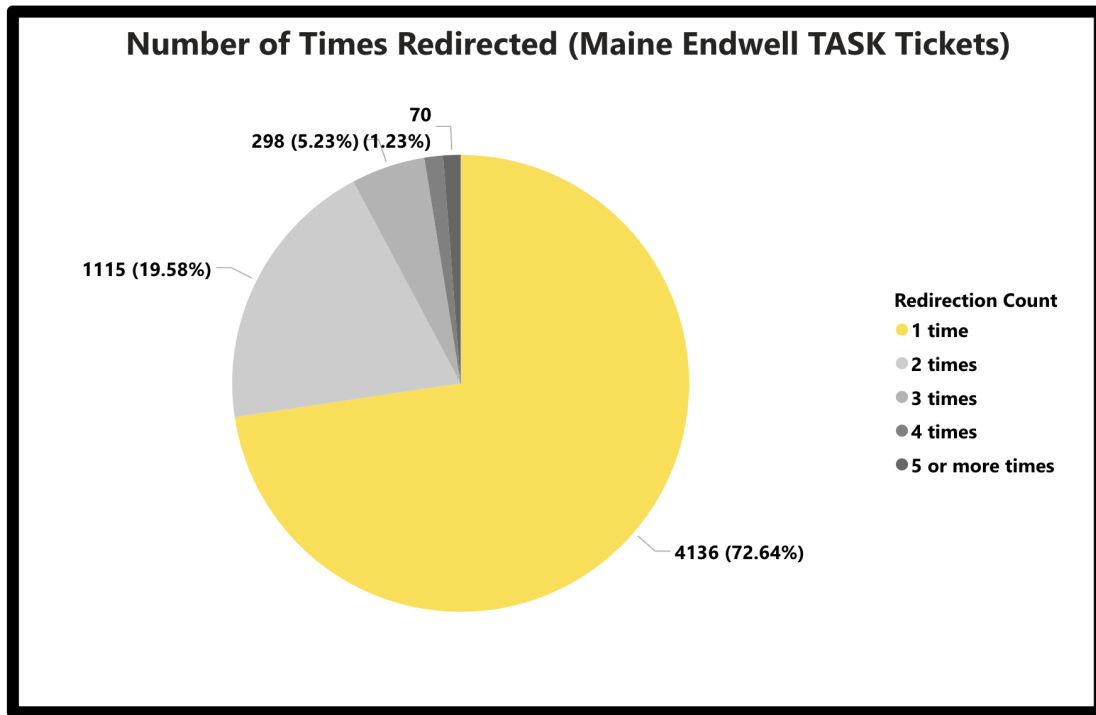


Figure A2. Pie chart adapted from that of the previous group with the current data for the TASK tickets for the Maine-Endwell district. As you can see, most tickets are redirected once.

Figure A2 focuses on TASK tickets, which show a notably stronger outcome. In this visualization, 72.64% of tickets were resolved after a single redirection, and 19.58% required two redirections. These results highlight a more streamlined routing process for TASK tickets compared to INC tickets.

Together, these visualizations reinforce the broader findings of this paper. While Maine-Endwell generally maintains efficient ticket handling, there remains room for refinement in initial routing, particularly for the incident-based tickets.

Appendix B

Additional Comparison: Ticket Resolution Time in Maine-Endwell

This appendix presents an analysis of the ticket resolution times in the Maine-Endwell District. The histograms below show how long tickets typically take to resolve, measured in hours, and offer insight into the range and frequency of resolution durations across all ticket types.

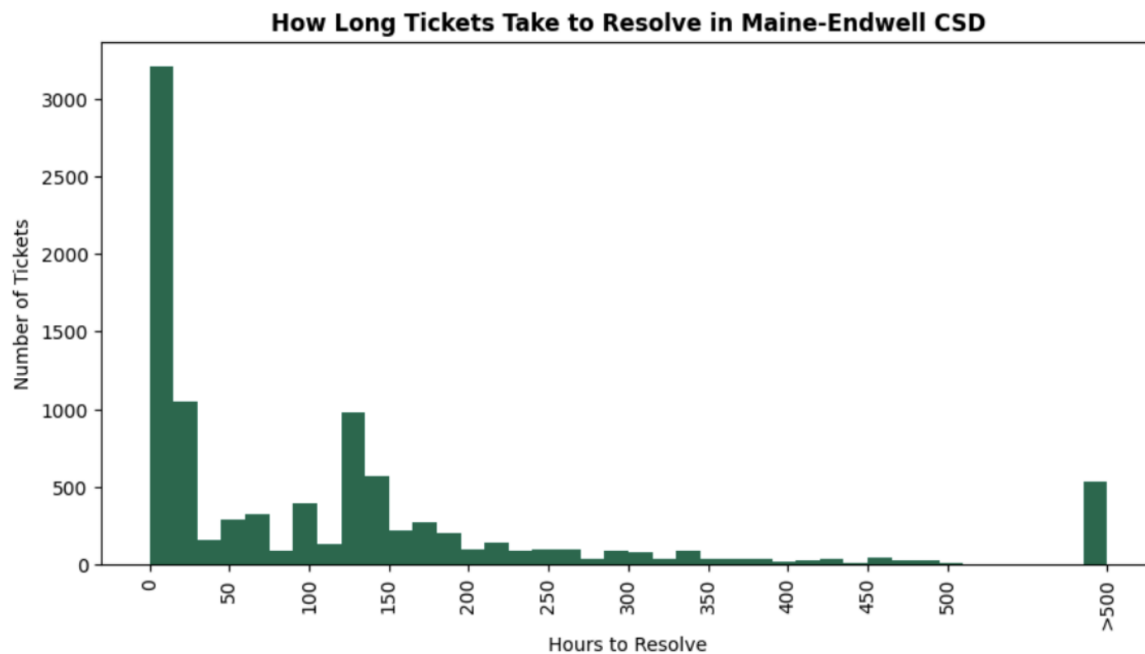


Figure B1. Histogram of how many tickets are resolved within certain time intervals in Maine-Endwell (Michals & Stewart).

Figure B1 is a histogram created by the previous group that provides a smooth, data-driven view of the frequency of ticket resolutions across hourly intervals. It shows a sharp spike in the 0-50 hour range, followed by a steep decline and random spikes in bins around 100-150 hours. This means that the majority of tickets are resolved within 1-2 days, but there are times when the tickets take longer. However, this is not as frequent of an occurrence. The distribution has a long tail, with a noticeable cluster of tickets that took more than 500 hours to resolve. This could be any tickets that had more than five transfers that caused the tickets to stay open for over 20 days.

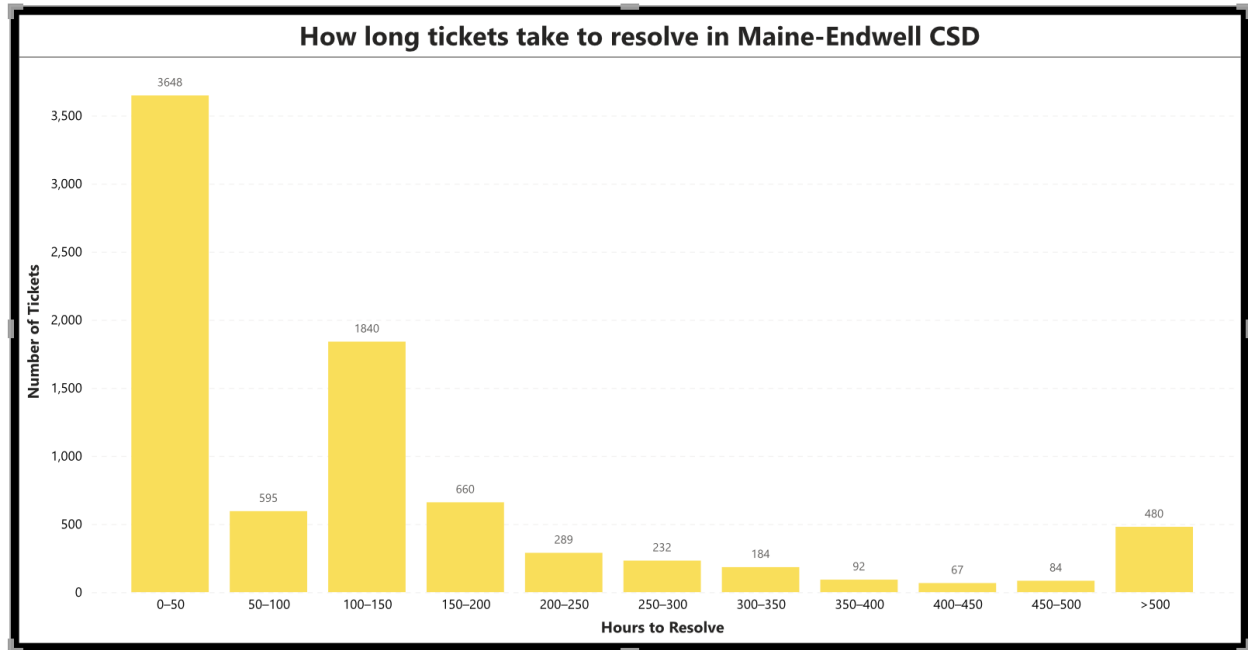


Figure B2. Histogram adapted from the previous group with current data that shows how many tickets were resolved within certain time intervals in Maine-Endwell.

Figure B2 presents the same trends as those of the previous group. This histogram was created using more recent data. However, the trends appear almost identical over the course of two academic years.